Mujdeh Hosein Ali - 40229067
AI Log Sprint 2

1. Task Objectives & Rationale
The primary goal was to construct a PostgreSQL relational database schema using Supabase to satisfy the Feature 2 and Feature 3 requirements outlined in the Sprint 2 guidelines.

Key constraints and system flows addressed:
- Enforcing the strict application state machine tracking workflow (`Applied`, `Under Review`, `Interview`, `Offered`, `Rejected`).
- Guaranteeing complete isolation and security of job seeker profiles using Row Level Security (RLS) until an application acts as an explicit bridge allowing the target recruiter access.
- Implementing indexing matrices to handle filtering parameters efficiently.
- Creating a diversified and resilient database seed script containing 20 jobs across distinct industries to facilitate frontend screen rendering tests.

2. Interaction Log (Prompts & Rationales)

Prompt 1: Initial Schema Generation
AI Rationale:** The model needed to read the state constraints from the provided project PDF guidelines (`Applied`, `Under Review`, `Interview`, `Offered`, `Rejected`) and translate them into a native PostgreSQL.

Prompt 2: Testing & Realistic Data Generation
User Intent:** Seed mock data matching the application criteria.
Prompt Submitted:
> "populate jobs table with a set of 20 jobs to be displayed on the users screen. use various jobs from various industries"
Followed by adjustment: "recruiter ids are unique to each recruiter that will sign up onthe website"
AI Rationale:** Hardcoding a single static text string for foreign keys fails database checks in runtime. The final iteration utilized an anonymous PL/pgSQL code block (`DO $$ ... END $$`) that scans the internal `auth.users` state table dynamically, extracts present system accounts, distributes rows through a modular looping formula, and automatically spins up safe default stubs if executed against a blank test slate.

Prompt 3: Relational & Security Audit
User Intent:** Align workflow authorization with business architecture.
Prompt Submitted:**
> "the job table is a list of job listings that job seekers can apply to and once applied, recruiters can check user's profiles. check the tables again"
AI Rationale:** Standard RLS locks profile vectors solely to the matching user account. To handle profile review pathways without turning off data isolation, the model drafted an active

multi-table cross-join query embedded within a dynamic conditional `EXISTS` check. Recruiters gain single-record reading privileges to profiles *if and only if* an intersection log binds that profile to their specific job posting.